Appendix C1 - South-west Corner Marine Park (western arm) Data Analysis Report

South-west Marine Parks Network

Table of contents

Data Analysis

Statistical methods and model selection results underpinning the natural values reported in Appendix A1.

Benthic ecosystem extent

Table C1.1: Habitat model selection results. Top generalised additive models (GAMs) for predicting the probability of occurrence of key habitats from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the most parsimonious model (fewest variables - shown in bold) within two units of the lowest AICc. Benthic samples are pooled across survey years, so survey year was not offered as a candidate predictor.

Response	Model	Δ AICc	ω AICc	R^2	EDF
Macroalgae	**detrended+aspect+Z**	**0**	**0.707**	**0.63419**	**6.95**
	detrended+roughness+aspect+Z	1.762	0.293	0.63419	7.81
Sand	**detrended+roughness+aspect+Z**	**0**	**1**	**0.50357**	**8.84**
Seagrass	**detrended+roughness+aspect+Z**	**0**	**0.997**	**0.38707**	**8.10**
Rock	**detrended+roughness+aspect+Z**	**0**	**1**	**0.18945**	**8.49**
Sessile invertebrates	**detrended+roughness+aspect+Z**	**0**	**1**	**0.39016**	**8.96**
Reef	**detrended+roughness+aspect+Z**	**0**	**1**	**0.45613**	**6.97**

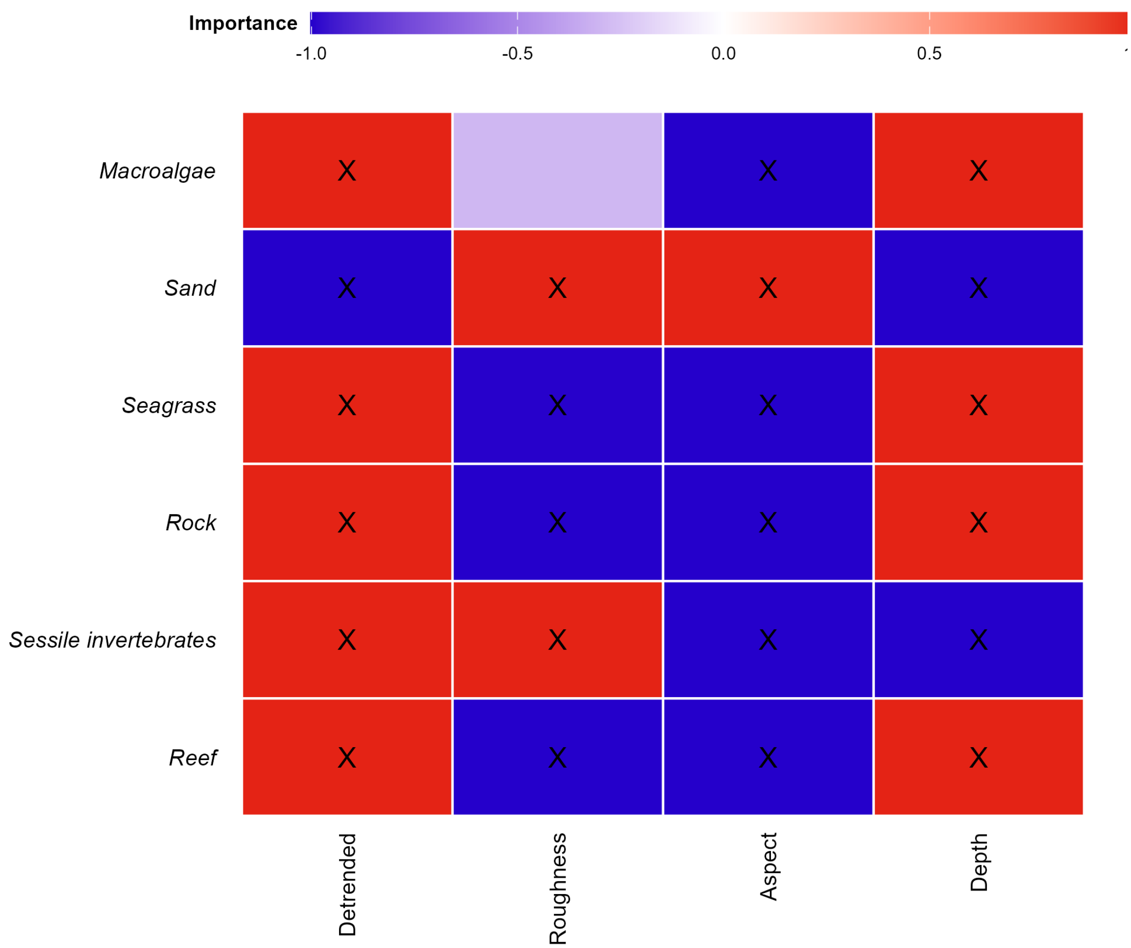

Figure C1.1: Variable importance scores for predicting the probability of occurrence of main habitat categories. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient.

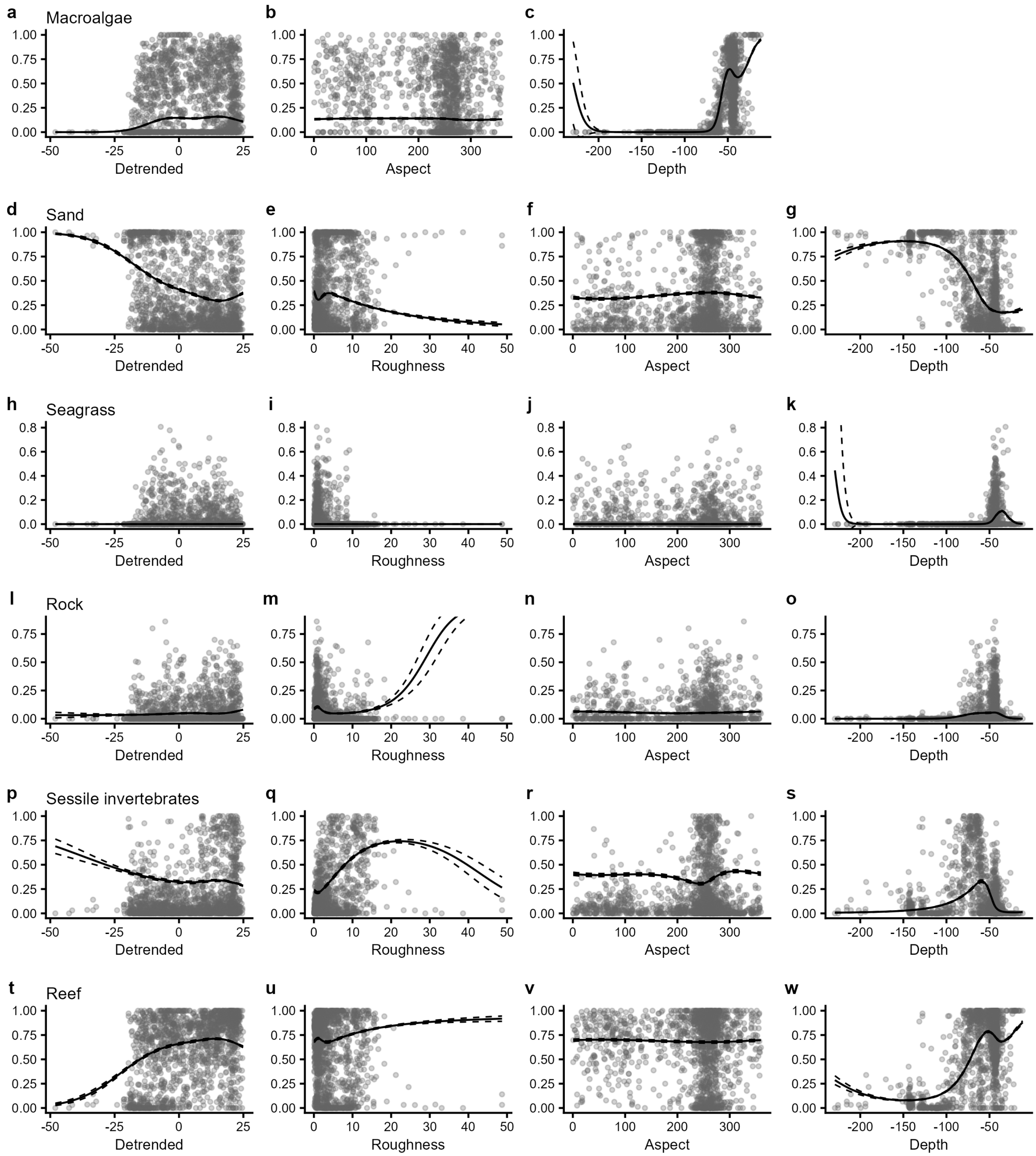

Figure C1.2: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types. Benthic samples from 2020, 2021, 2023, 2024 and 2025 are pooled, because the campaigns sample different areas and a survey-year term would read as a spatial difference rather than change through time. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data.

Demersal fish assemblage

Table C1.2: Demersal fish model selection results. Top generalised additive models (GAMs) for predicting the species and abundance distribution of metrics of interest from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the best model (fewest variables - shown in bold) within two units of the lowest AICc. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Response	Model	Δ AICc	ω AICc	AICc	R^2	EDF
Species richness	roughness+aspect+reef+year+status	0	0.672	3504.19	0.45352	8.24
	roughness+reef+year+status	**1.435**	**0.328**	**3505.63**	**0.44939**	**6.99**
Total abundance	**roughness+aspect+reef+year+status**	**0**	**0.726**	**6313.92**	**0.35566**	**8.47**
Large Reef Fish Index*	roughness+aspect+reef+year+status	0	0.545	10803.19	0.14721	8.19
	roughness+reef+year+status	**1.031**	**0.325**	**10804.22**	**0.14192**	**6.91**
Reef Fish Thermal Index	**roughness+reef+year+status**	**0**	**0.339**	**1615.67**	**0.06865**	**6.15**
	roughness+aspect+reef+year+status	0.001	0.339	1615.67	0.06865	7.15

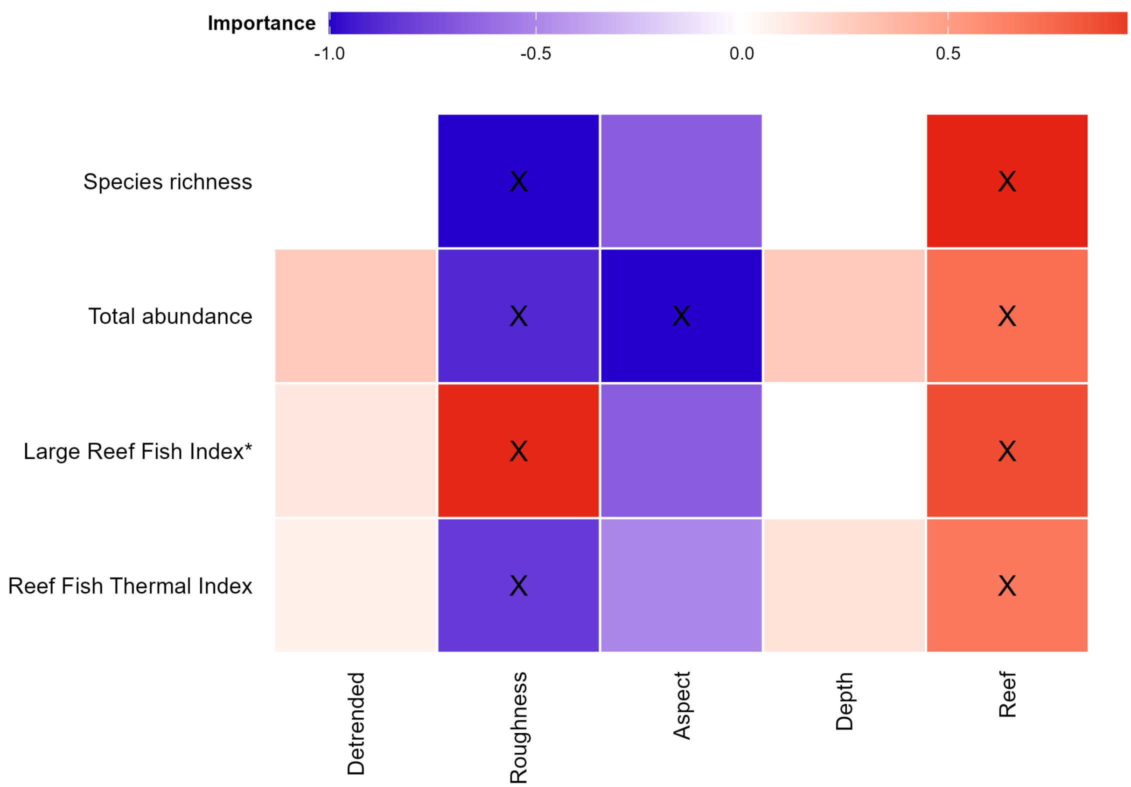

Figure C1.3: Variable importance scores for predicting the probability of demersal fish metrics. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

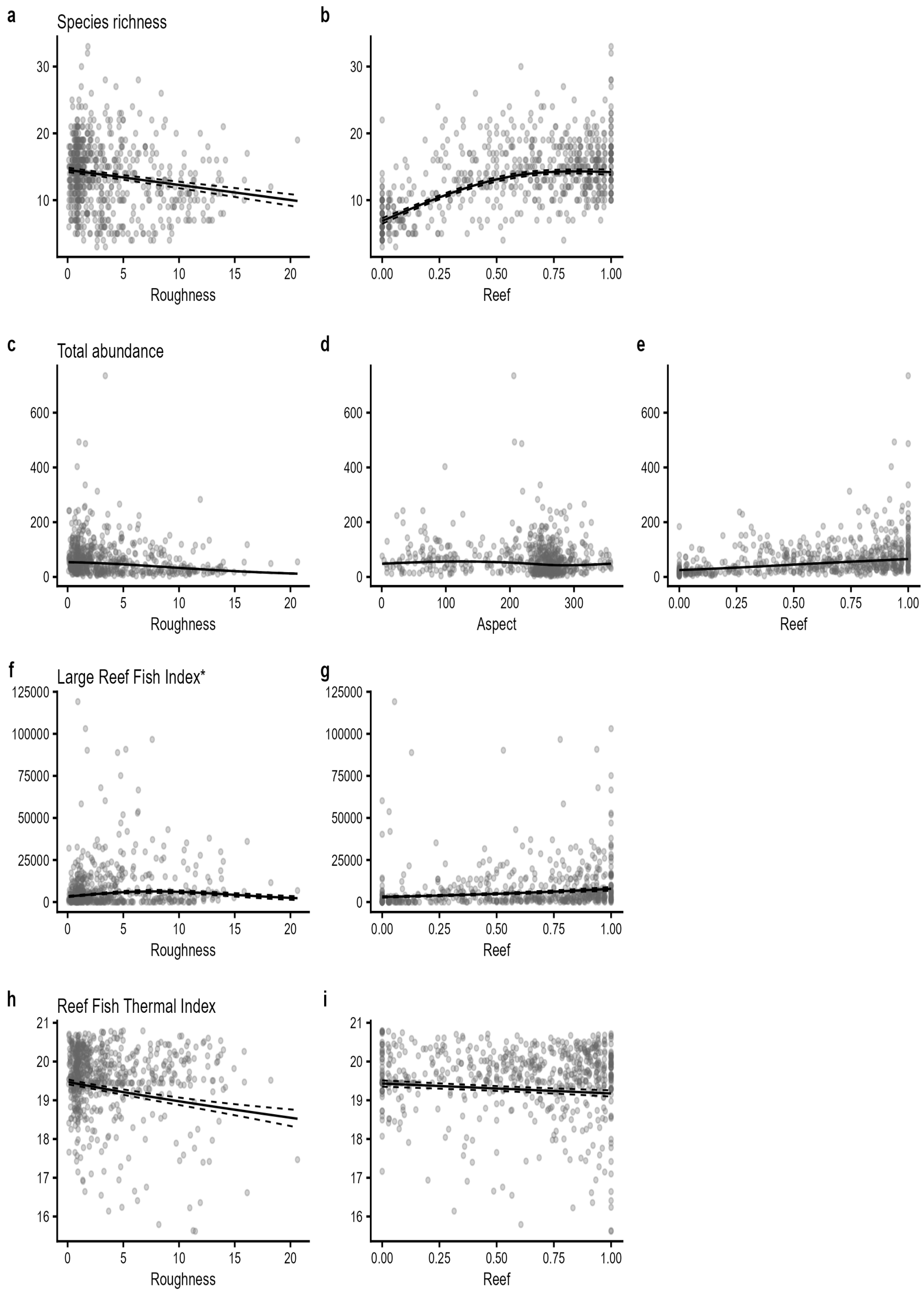

Figure C1.4: Plots of the best model found to predict the probability of occurrence of key fish groups. Individual panes display the predicted relationships between model covariates and each demersal fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value, with year held at 2020 and zone status held at areas open to fishing. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years and zones. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Significance of year and zone status

Table C1.3: Significance of the year and zone status terms in each demersal fish GAMM. Estimate, standard error (SE), test statistic and p-value are taken from the parametric coefficient table of each model summary, one row per factor-level contrast (e.g. a survey year against the reference year, or No-Take against Fished).

Response	Term	Estimate	SE	Statistic	p-value
Species richness	Zone status: No-Take vs Fished	0.054	0.307	0.178	0.859
	Year: 2024 vs 2020	3.277	0.374	8.754	<0.001
	Year: 2025 vs 2020	0.524	0.418	1.252	0.211
Total abundance	Zone status: No-Take vs Fished	0.053	0.010	5.329	<0.001
	Year: 2024 vs 2020	0.527	0.012	42.525	<0.001
	Year: 2025 vs 2020	0.612	0.013	47.134	<0.001
Large Reef Fish Index*	Zone status: No-Take vs Fished	0.294	0.101	2.923	0.004
	Year: 2024 vs 2020	0.734	0.122	5.998	<0.001
	Year: 2025 vs 2020	0.346	0.137	2.522	0.012
Reef Fish Thermal Index	Zone status: No-Take vs Fished	0.124	0.067	1.841	0.066
	Year: 2024 vs 2020	0.462	0.082	5.617	<0.001
	Year: 2025 vs 2020	0.319	0.092	3.468	<0.001

Table C1.4: Zone status test for each demersal fish metric, laid out by survey year. Zone status (No-Take vs Fished) is fitted as a single term across all years (no year by status interaction), so the status estimate, SE and p-value are the same in every row for a given metric; predicted values and their SE are shown per year, with other model covariates held at their mean.

Response	Year	Fished (predicted)	Fished (SE)	No-Take (predicted)	No-Take (SE)	Status estimate	Status SE	Status p-value
Species richness	2020	13.713	0.334	13.768	0.357	0.054	0.307	0.859
	2024	16.990	0.362	17.045	0.392	0.054	0.307	0.859
	2025	14.237	0.431	14.292	0.468	0.054	0.307	0.859
Total abundance	2020	48.965	0.800	51.618	0.842	0.053	0.010	<0.001
	2024	82.949	1.226	87.444	1.302	0.053	0.010	<0.001
	2025	90.252	1.500	95.143	1.619	0.053	0.010	<0.001
Large Reef Fish Index*	2020	5302.069	608.551	7115.389	752.770	0.294	0.101	0.004
	2024	11051.046	1176.538	14830.530	1558.748	0.294	0.101	0.004
	2025	7497.189	925.291	10061.245	1234.366	0.294	0.101	0.004
Reef Fish Thermal Index	2020	19.281	0.063	19.404	0.062	0.124	0.067	0.066
	2024	19.743	0.071	19.866	0.073	0.124	0.067	0.066
	2025	19.600	0.080	19.723	0.083	0.124	0.067	0.066